

AWS IAM Customer Managed Policies

To Begin with the Lab

Summary of the Lab

In this lab, **AWS IAM** customer managed policies were used to solve a fine-grained permission problem that **AWS Managed Policies** could not handle. Two IAM users, **EC2User** and **VPCUser**, were created first so the lab could compare broad AWS-managed access with custom user-defined access. The **EC2User** was then given read-only access first, and later a customer managed policy was created to allow only selected **Amazon EC2** instances to be started, stopped, or terminated. The lab also demonstrated policy evaluation rules, including default deny, explicit allow, and explicit deny overriding allow. Finally, the policy simulator was used to verify that the custom policy behaved correctly and that policy size limits still applied.

Understanding Customer Managed Policies in AWS IAM

A **customer managed policy** is a JSON-based policy created and managed by the user instead of **AWS**. It exists because **AWS Managed Policies** are easy to attach but usually too broad for situations where you need exact control over which resource a user can manage. With a customer managed policy, you can grant permissions for only specific resources, such as only two **Amazon EC2** instances out of three, and you can update or version the policy when requirements change. This makes it useful when a business wants tighter security, smaller blast radius, and permission control that matches the job role exactly.

Example:

An IAM user named **EC2User** needs to manage only server1 and server2, but must not touch server3. A custom policy is created with `ec2:StartInstances`, `ec2:StopInstances`, and `ec2:TerminateInstances` allowed only for the ARNs of server1 and server2. Later, an explicit deny policy is attached for server2 so even an existing allow cannot override it.

Lab Steps

Step 1 – Create the IAM Users

- Sign in to the AWS Management Console.
- Open **IAM**.
- Click **Users**.

Identity and Access Management (IAM)

Search IAM

Dashboard

▼ Access Management

- Roles
- Policies
- IAM users**
- IAM user groups
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

▼ Access reports

- Access Analyzer
- Resource analysis
- Unused access
- Analyzer settings
- Credential report
- Organization activity
- Service control policies
- Resource control policies

IAM Dashboard

Security recommendations

- Root user has MFA: Having multi-factor authentication (MFA) for the root user improves security for this account.
- Add MFA for yourself: Add multi-factor authentication (MFA) for yourself to improve security for this account.
- Your user, cf-shashank1, does not have any active access keys that have been unused for more than a year. Deactivating or deleting unused access keys improves security.

IAM resources

Resources in this AWS Account

You are approaching your quota of available IAM resources. [Learn more](#)

User groups	Users	Roles	Policies	Identity providers
42	8	415	1132 368 available	1

What's new

Updates for features in IAM

AWS Account

Account ID: [redacted]

Account Alias: [redacted]

Sign-in URL for IAM users in this account: [https://\[redacted\].signin.aws.amazon.com/console](https://[redacted].signin.aws.amazon.com/console)

Quick Links

[My security credentials](#)

Manage your access keys, multi-factor authentication (MFA) and other credentials.

Tools

[Policy simulator](#)

The simulator evaluates the policies that you choose and determines the effective permissions for each of the actions that you specify.

- Click **Create user**.

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Managing human user access account by account? There's a better way.

Streamline human access to AWS and cloud apps when you enable Identity Center. [Learn more](#)

[Watch how it works](#)

- One-time set up for workforce user access
- Centrally manage access to multiple AWS accounts
- Provide access centrally to the cloud applications your workforce uses
- All with one-click access through a simple web portal

IAM users (6)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Search

☐	User name	Path	Group	Last activity	MFA	Password age	Console last sign-in	Access key ID
☐	[redacted]	/	0	10 hours ago	-	6 days	16 hours ago	Active - [redacted]
☐	[redacted]	/	0	1 hour ago	-	16 days	1 hour ago	-
☐	[redacted]	/	0	21 hours ago	-	5 days	21 hours ago	-
☐	[redacted]	/	1	5 days ago	-	5 days	5 days ago	-
☐	[redacted]	/	1	18 hours ago	-	19 hours	19 hours ago	Active - [redacted]
☐	[redacted]	/	1	10 hours ago	-	10 hours	-	-

[Delete](#) [Create user](#)

- Create the first user as **EC2User**.
- Enable AWS Console access.
- Set a console password.
- Leave the user without permissions.

User details

User name
EC2User

☒ Provide user access to the AWS Management Console *optional*

Are you providing console access to a person?

User type

☐ Specify a user in Identity Center - Recommended

☒ I want to create an IAM user

Console password

☐ Autogenerated password

☒ Custom password

☐ Show password

☒ Users must create a new password at next sign-in - Recommended

- Click **Create user**.
- Create the second user as **VPCUser** with the same steps as first user.

Step 2 – Demonstrate AWS Managed Policy Behavior

- Open the **EC2User** account in a separate browser session.
- Attach **AmazonEC2ReadOnlyAccess** to **EC2User**.

Add permissions

Permissions options

☐ Add user to group

☐ Copy permissions

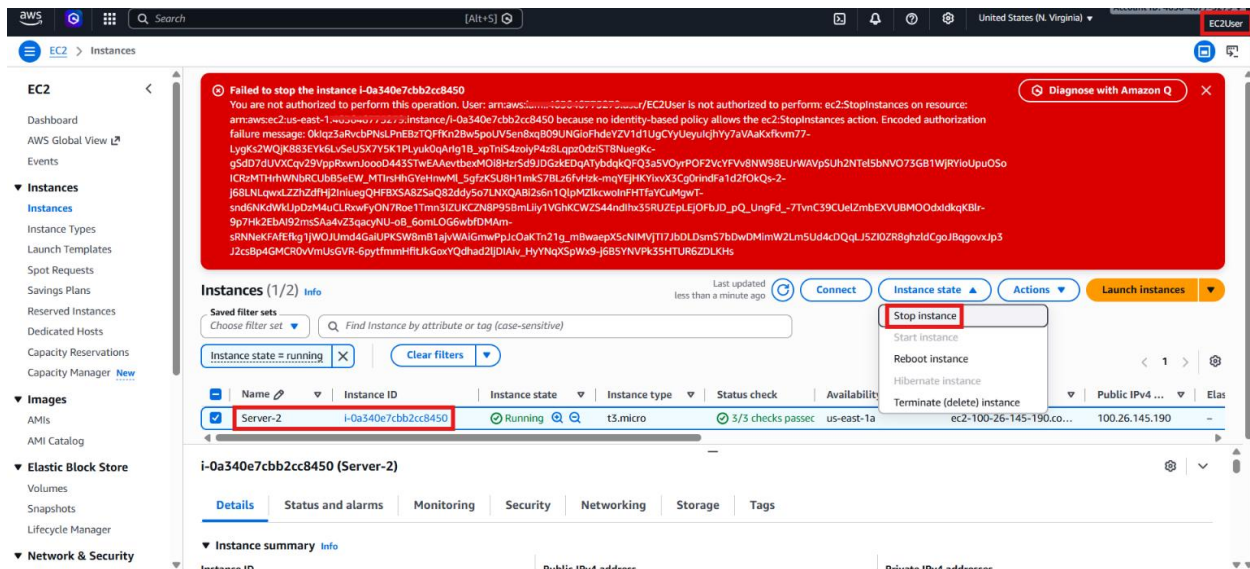
☒ Attach policies directly

Permissions policies (1/2638)

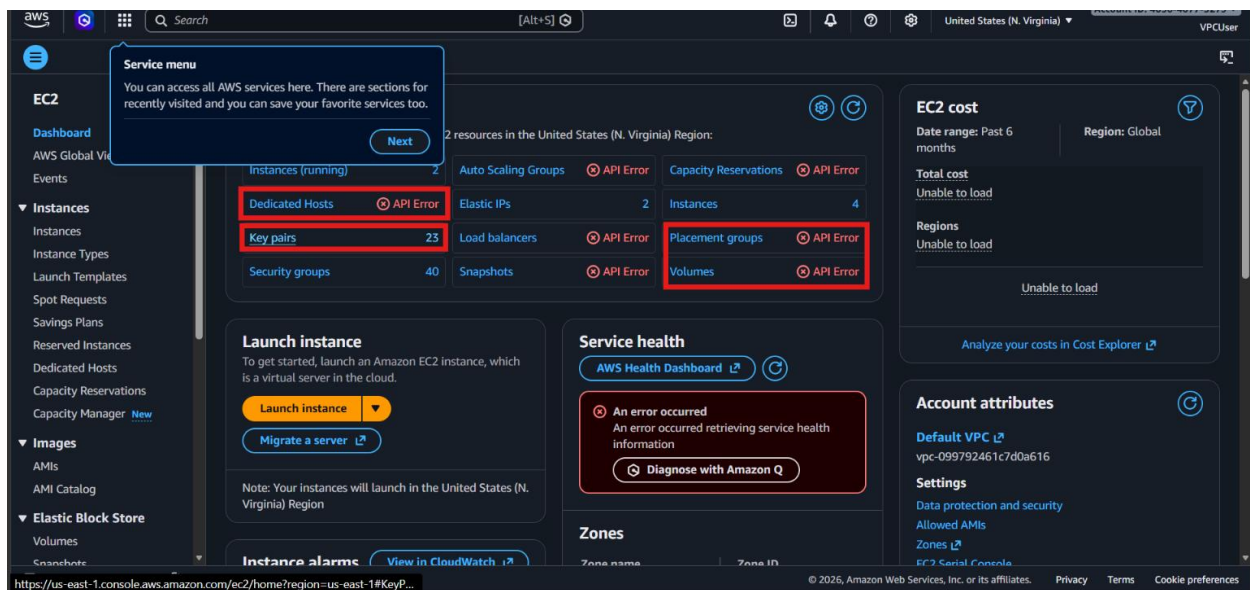
Policy name	Type	Attached entities
☒ AmazonEC2ReadOnlyAccess	AWS managed	0
☐ AWSLambdaManagedEC2ResourceOperator	AWS managed	0
☐ DeclarativePoliciesEC2Report	AWS managed	0

Cancel **Next**

- Refresh the console.
- Verify that **EC2User** can view **Amazon EC2** resources but cannot start or stop instances.

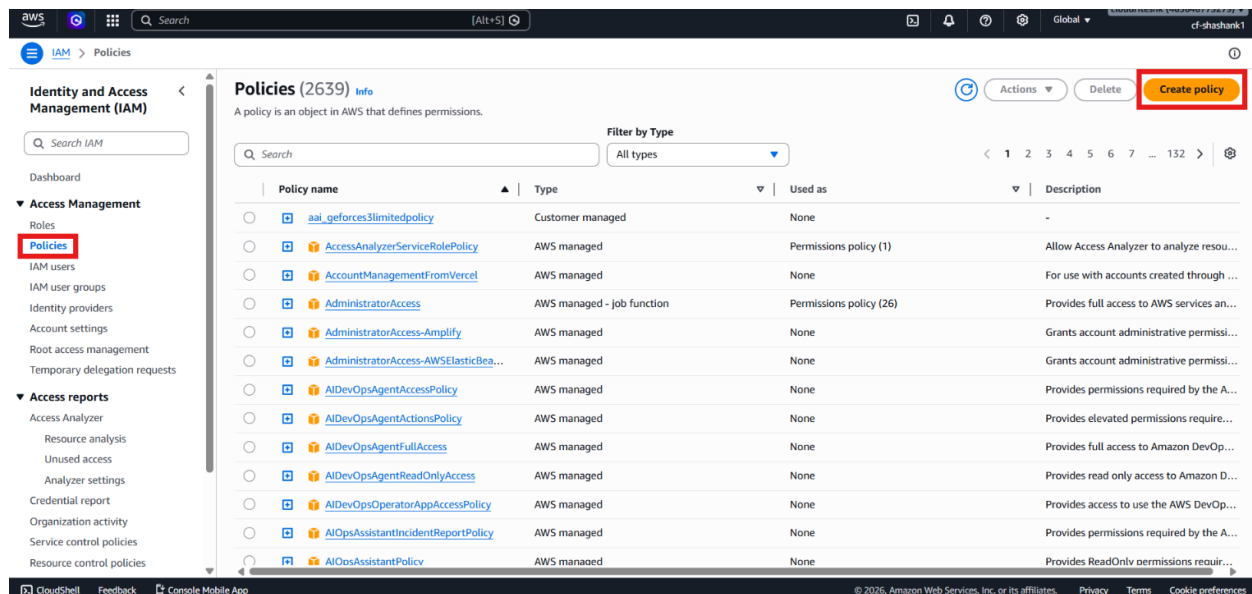


- Open the **VPCUser** account in another browser session.
- Attach **AmazonVPCFullAccess** to **VPCUser**.
- Refresh the console.
- Verify that **VPCUser** can manage **Amazon VPC** resources but does not have full **Amazon EC2** control.



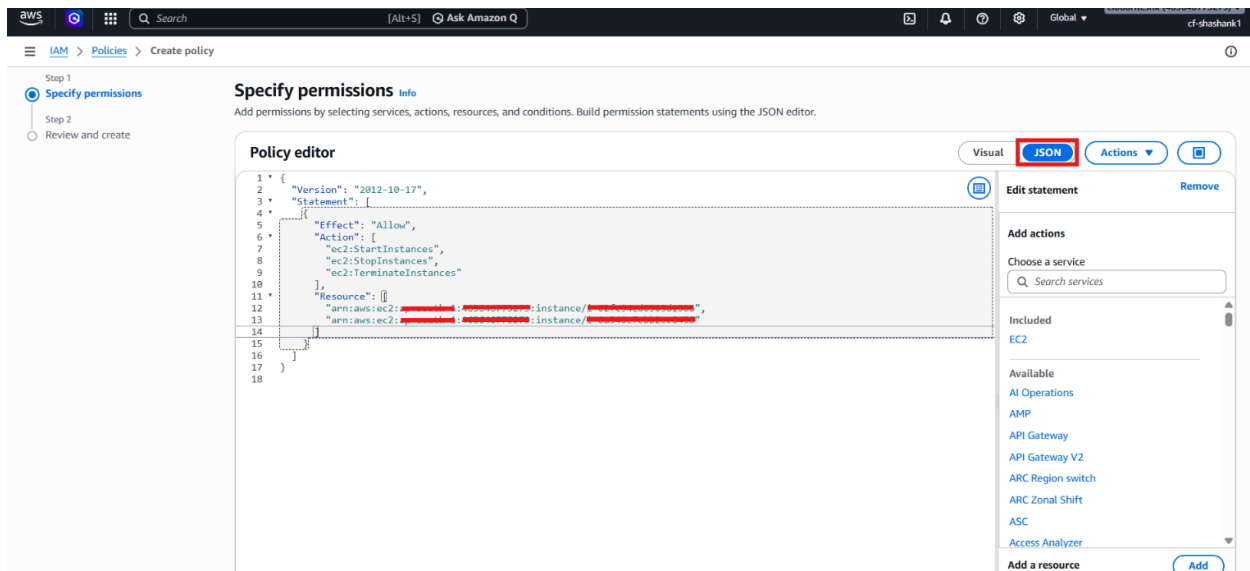
Step 3 – Create the Customer Managed Policy

- Go to **IAM** → **Policies**.
- Click **Create policy**.

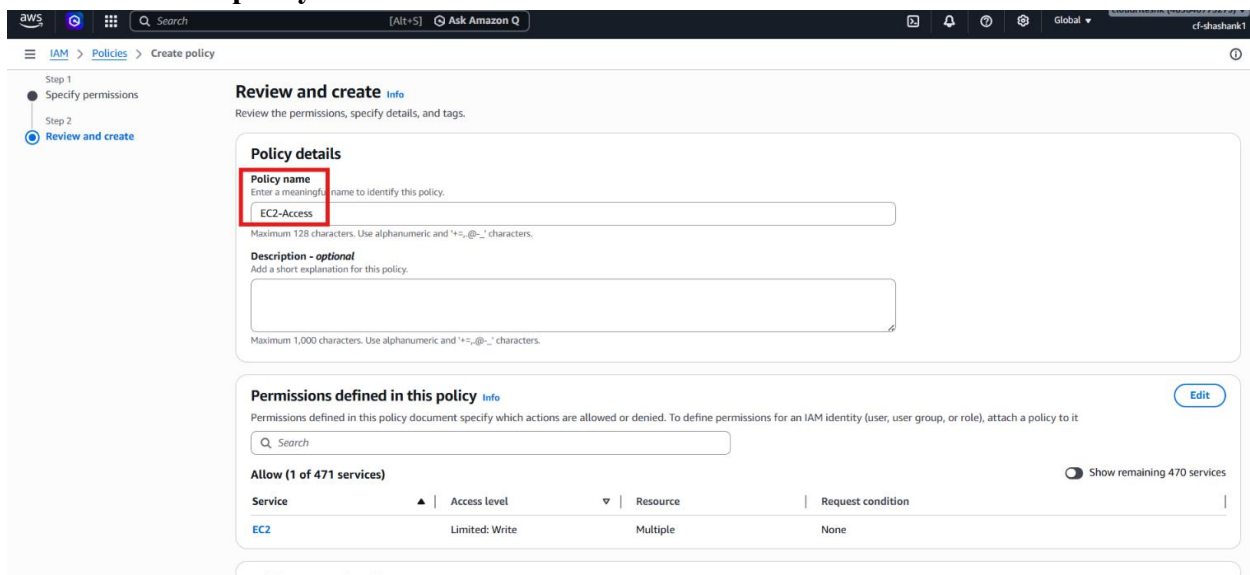


- Choose the **JSON** editor.
- Add a policy that allows only `ec2:StartInstances`, `ec2:StopInstances`, and `ec2:TerminateInstances`.
- Restrict the policy to the ARNs of only the two allowed EC2 instances.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "ec2:StartInstances",
        "ec2:StopInstances",
        "ec2:TerminateInstances"
      ],
      "Resource": [
        "arn:aws:ec2:<region>:<account-id>:instance/<instance-id-1>",
        "arn:aws:ec2:<region>:<account-id>:instance/<instance-id-2>"
      ]
    }
  ]
}
```

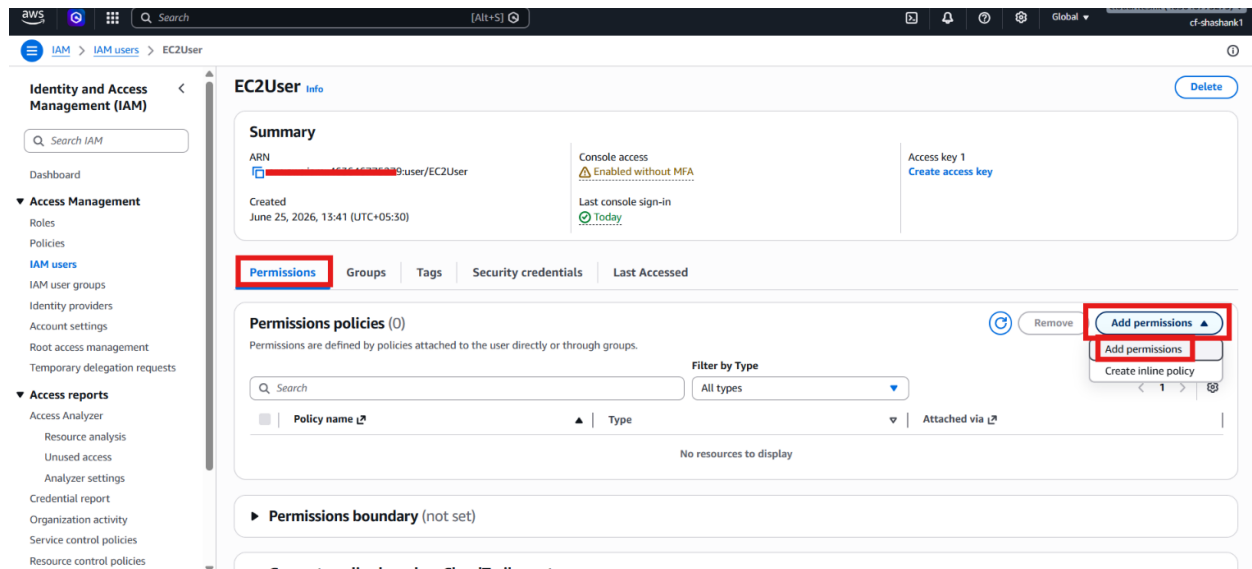


- Name the policy.
- Click **Create policy**.

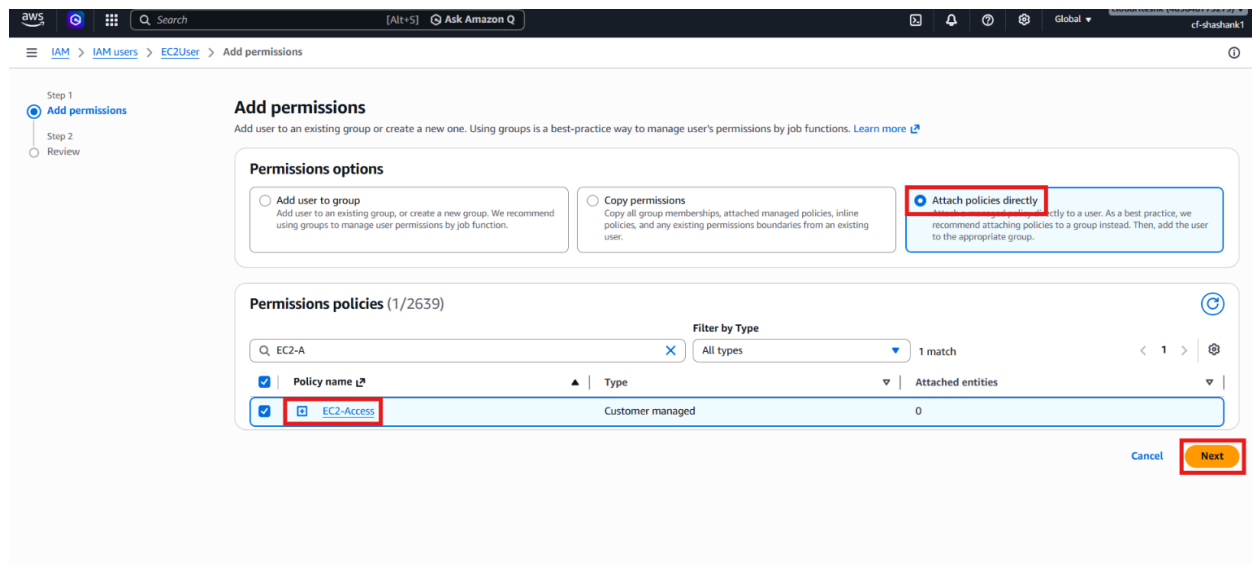


Step 4 – Attach the Policy to EC2User

- Open the EC2User permissions page.
- Click **Add permissions**.



- Choose **Attach policies directly**.
- Select the customer managed policy you created.
- Click **Next**.



- Click **Add permissions**.
- Refresh the EC2 console session for **EC2User**.

EC2User Info Delete

Summary

ARN: [arn:aws:iam::163646779279:user/EC2User](#)

Created: June 25, 2026, 13:41 (UTC+05:30)

Console access: Enabled without MFA

Last console sign-in: Today

Access key 1: [Create access key](#)

Permissions policies (2) Remove Add permissions

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type: All types

Policy name	Type	Attached via
AmazonEC2ReadOnlyAccess	AWS managed	Directly
EC2-Access	Customer managed	Directly

- Verify that **EC2User** can manage only the allowed instances.
- Confirm that **Server-1** and **Server-2** can be started, stopped, or terminated.

Instances (2/3) Info Connect Instance state Actions Launch instances

Instance state = running Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic ...
☒ Server-2	i-0a340e7cbb2cc8450	Stopping	t3.micro	-	us-east-1a	ec2-100-26-145-190.co...	100.26.145.190	-
☒ Server-1	i-02fc94ed0903d1506	Stopping	t3.micro	-	us-east-1a	ec2-100-53-190-203.co...	100.53.190.203	-
☐ Server-3	i-013773099d1b54be6	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-34-207-144-94.co...	34.207.144.94	-

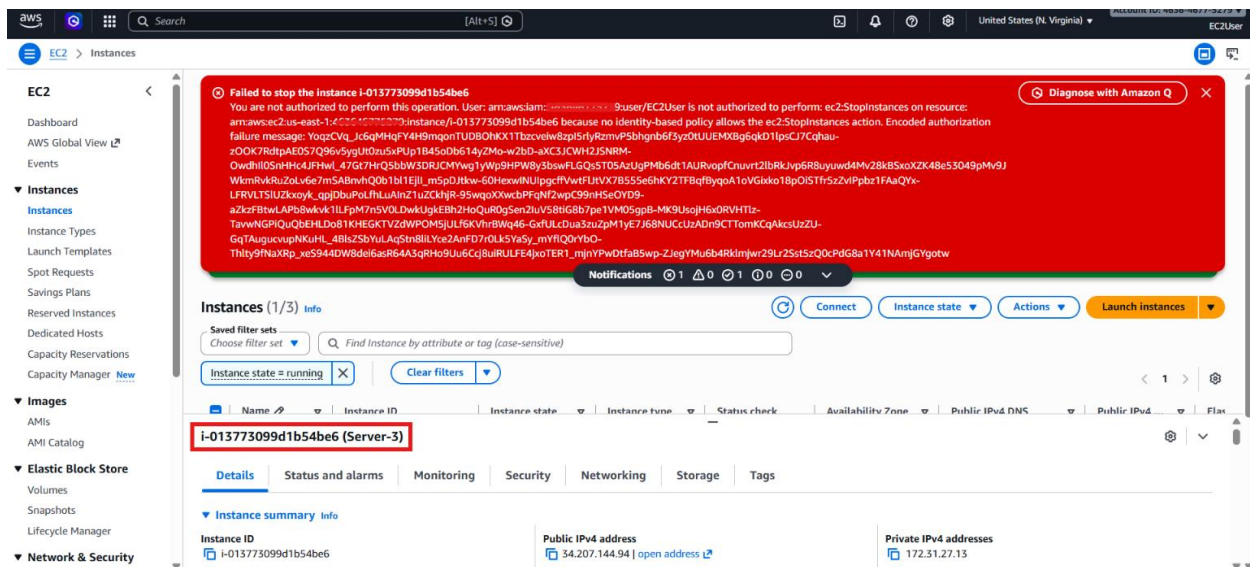
2 instances selected

Monitoring

☐ Alarm recommendations Investigate with AI - new 1h 3h 12h 1d 3d 1w Custom UTC timezone Explore related

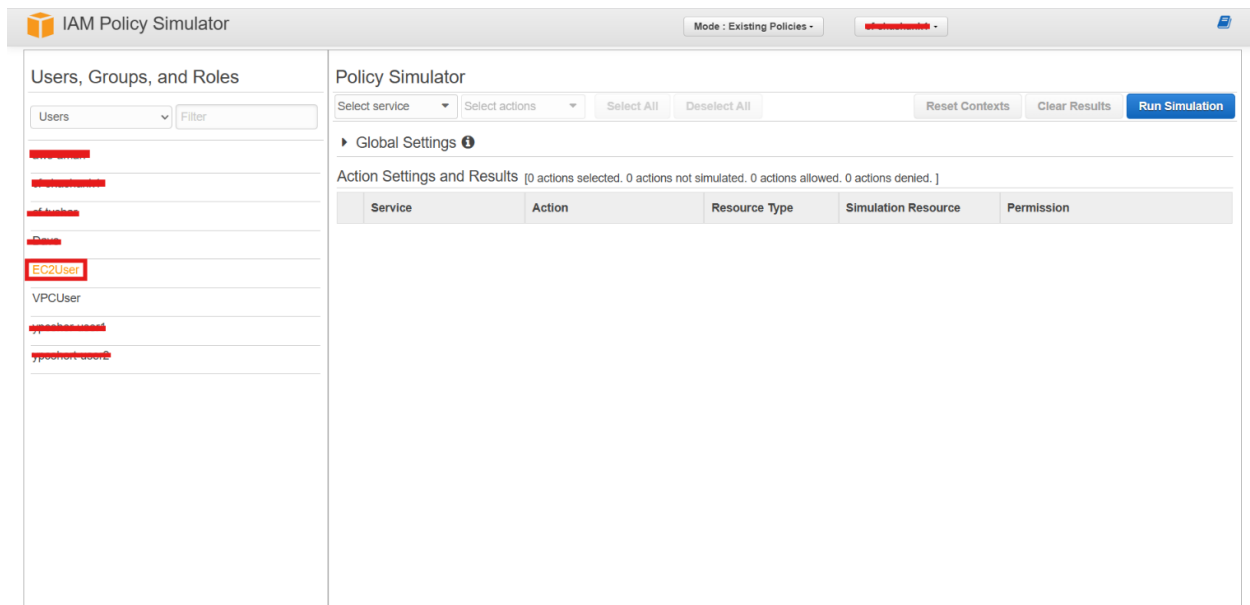
CPU utilization (%) **Network in (bytes)** **Network out (bytes)** **Network packets in (count)**

- Confirm that **Server-3** remains inaccessible for those actions.

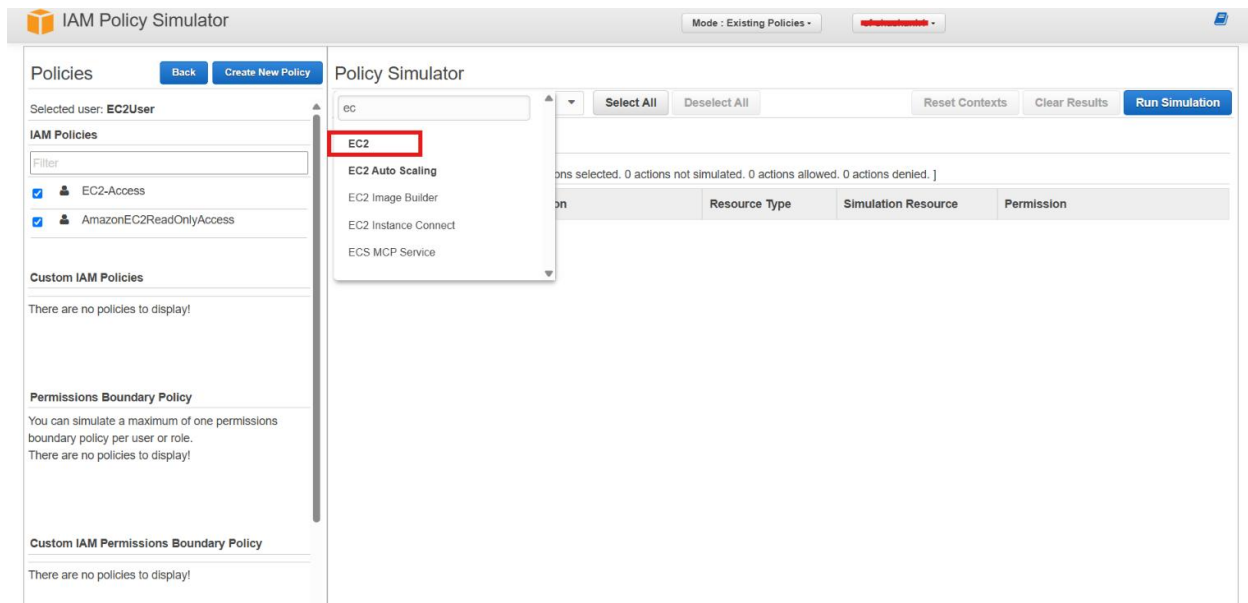


Step 5 – Test with the IAM Policy Simulator

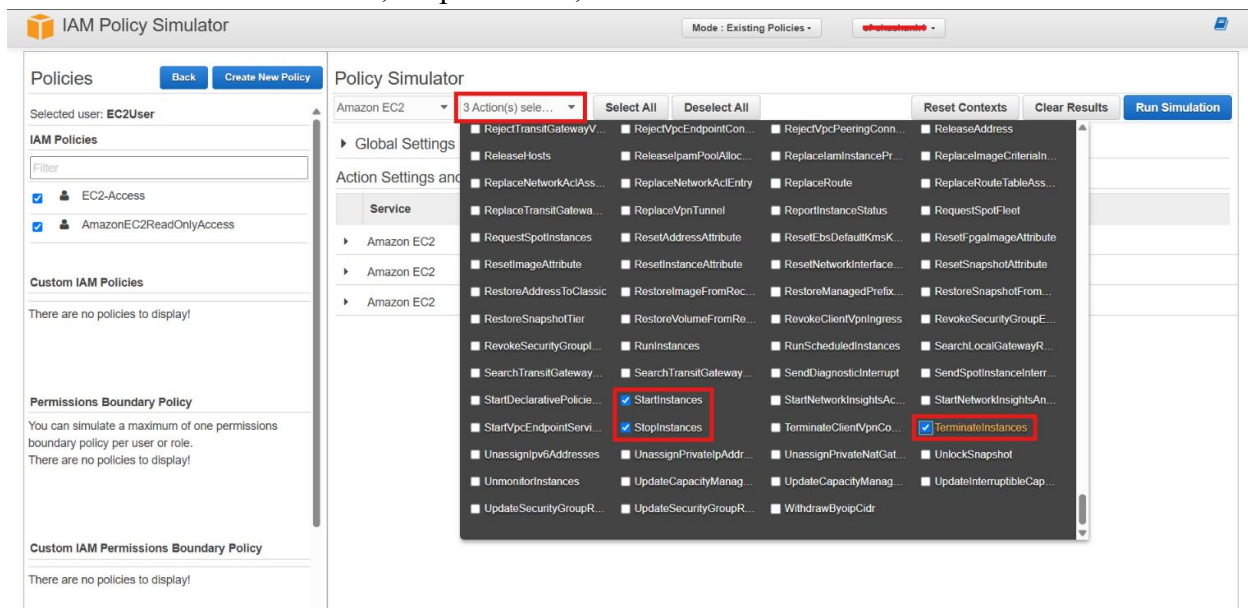
- Open the **IAM Policy Simulator**.
- You can open it up here: <https://policysim.aws.amazon.com/>
- Select **EC2User**.



- Choose **Amazon EC2** as the service.



- Select Test StartInstances, StopInstances, and TerminateInstances actions and then test them.



- Verify that the simulator allows only the approved instance ARNs.
- Click on **Run Simulation**.
- Verify that the simulator denies the action when the policy does not allow all instances.

The screenshot shows the IAM Policy Simulator interface. On the left, under 'Policies', the selected user is 'EC2User'. Under 'IAM Policies', 'EC2-Access' and 'AmazonEC2ReadOnlyAccess' are selected. The 'Policy Simulator' section on the right shows 'Amazon EC2' as the service and '3 Action(s) selected'. The 'Run Simulation' button is highlighted in red. Below, the 'Action Settings and Results' table shows three actions: 'StartInstances', 'StopInstances', and 'TerminateInstances', all of which are denied.

Service	Action	Resource Type	Simulation Resource	Permission
Amazon EC2	StartInstances	Instance	*	denied Implicitly denied (no matching st...
Amazon EC2	StopInstances	Instance	*	denied Implicitly denied (no matching st...
Amazon EC2	TerminateInstances	Instance	*	denied Implicitly denied (no matching st...

- Confirm the policy behaves as expected before applying it in production.

Verification

- **EC2User** and **VPCUser** were created successfully.
- **AWS Managed Policies** demonstrated broad default permissions.
- The customer managed policy granted access only to selected **Amazon EC2** instances.
- The policy simulator confirmed the expected behavior.